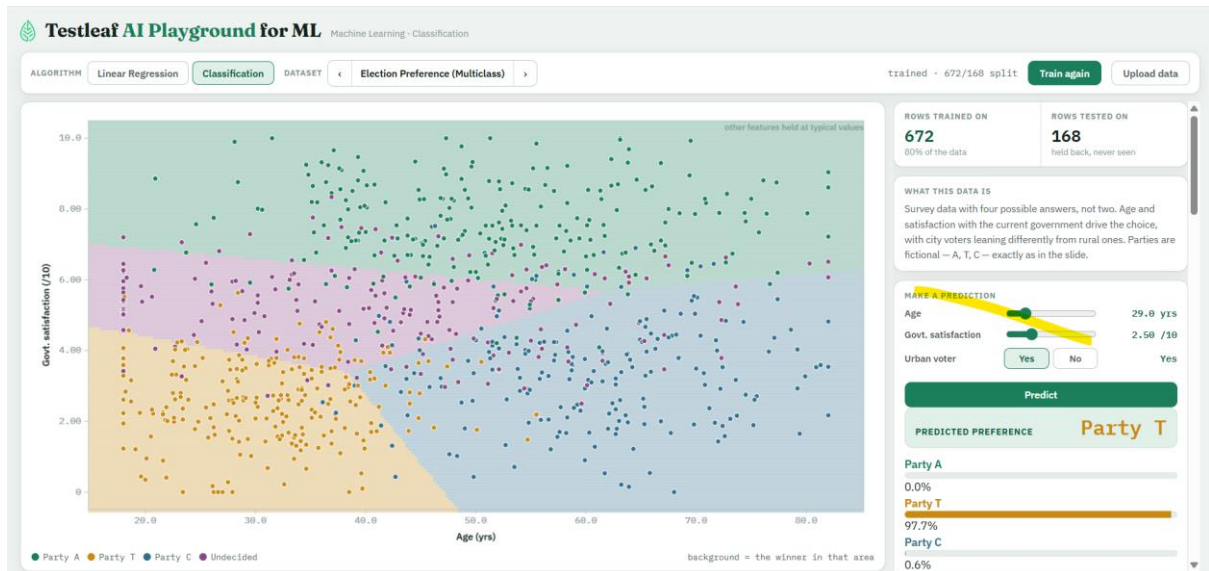


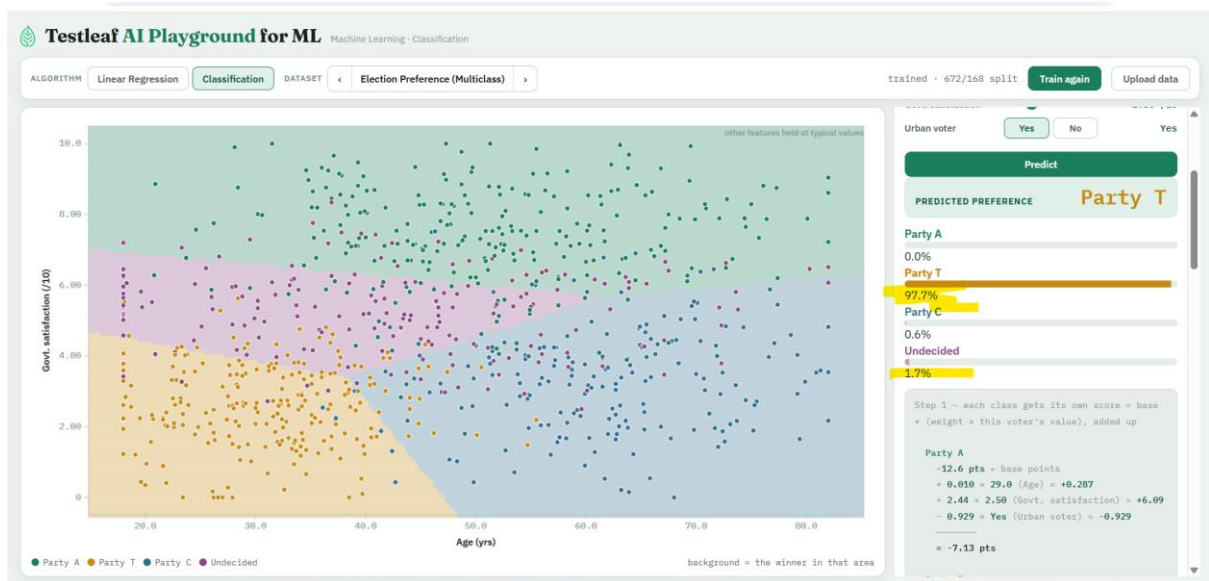
1) Classification Algorithm

1) Age <30 and Govt satisfaction less , urban

2) Age >50 and Govt satisfaction less , non urban validate the percentage.

1.





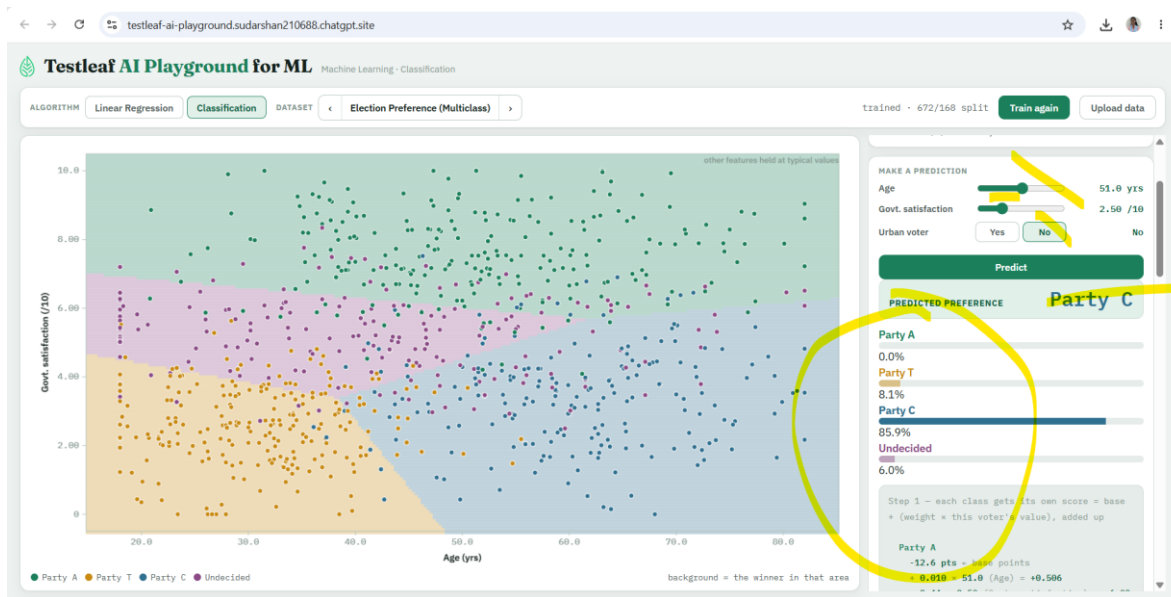
Step 2b – each class's share = its own e^{score} ÷ that same total (96.7)

Party A	$0.001 \div 96.7 = 0.0\%$
Party T	$94.4 \div 96.7 = 97.7\%$
Party C	$0.593 \div 96.7 = 0.6\%$
Undecided	$1.64 \div 96.7 = 1.7\%$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

Party T - 97.7%

2.



Party C with 85.9

Party A

$$\begin{aligned} & -12.6 \text{ pts} \leftarrow \text{base points} \\ & + 0.010 \times 51.0 \text{ (Age)} = +0.506 \\ & + 2.44 \times 2.50 \text{ (Govt. satisfaction)} = +6.09 \\ & - 0.929 \times \text{No (Urban voter)} = +0 \\ & \hline & = -5.98 \text{ pts} \end{aligned}$$

Party T

$$\begin{aligned} & 10.2 \text{ pts} \leftarrow \text{base points} \\ & - 0.141 \times 51.0 \text{ (Age)} = -7.19 \\ & - 1.10 \times 2.50 \text{ (Govt. satisfaction)} = -2.75 \\ & + 1.23 \times \text{No (Urban voter)} = +0 \\ & \hline & = 0.222 \text{ pts} \end{aligned}$$

Party C

$$\begin{aligned} & -0.536 \text{ pts} \leftarrow \text{base points} \\ & + 0.086 \times 51.0 \text{ (Age)} = +4.40 \\ & - 0.513 \times 2.50 \text{ (Govt. satisfaction)} = \\ & -1.28 \\ & - 1.21 \times \text{No (Urban voter)} = +0 \\ & \hline & = 2.58 \text{ pts} \end{aligned}$$

Undecided

$$\begin{aligned} & -0.019 \text{ pts} \leftarrow \text{base points} \\ & - 0.038 \times 51.0 \text{ (Age)} = -1.94 \\ & + 0.748 \times 2.50 \text{ (Govt. satisfaction)} = \\ & +1.87 \end{aligned}$$

= -0.085 pts

highest score wins → Party C

Step 2a – convert each score to e^{score}
(always positive)

Party A $e^{-5.98} = 0.003$

Party T $e^{0.222} = 1.25$

Party C $e^{2.58} = 13.2$

Undecided $e^{-0.085} = 0.919$

Total (add all 4 up) = 15.4

Step 2b – each class's share = its own e^{score}
÷ that same total (15.4)

Party A $0.003 \div 15.4 = 0.0\%$

Party T $1.25 \div 15.4 = 8.1\%$

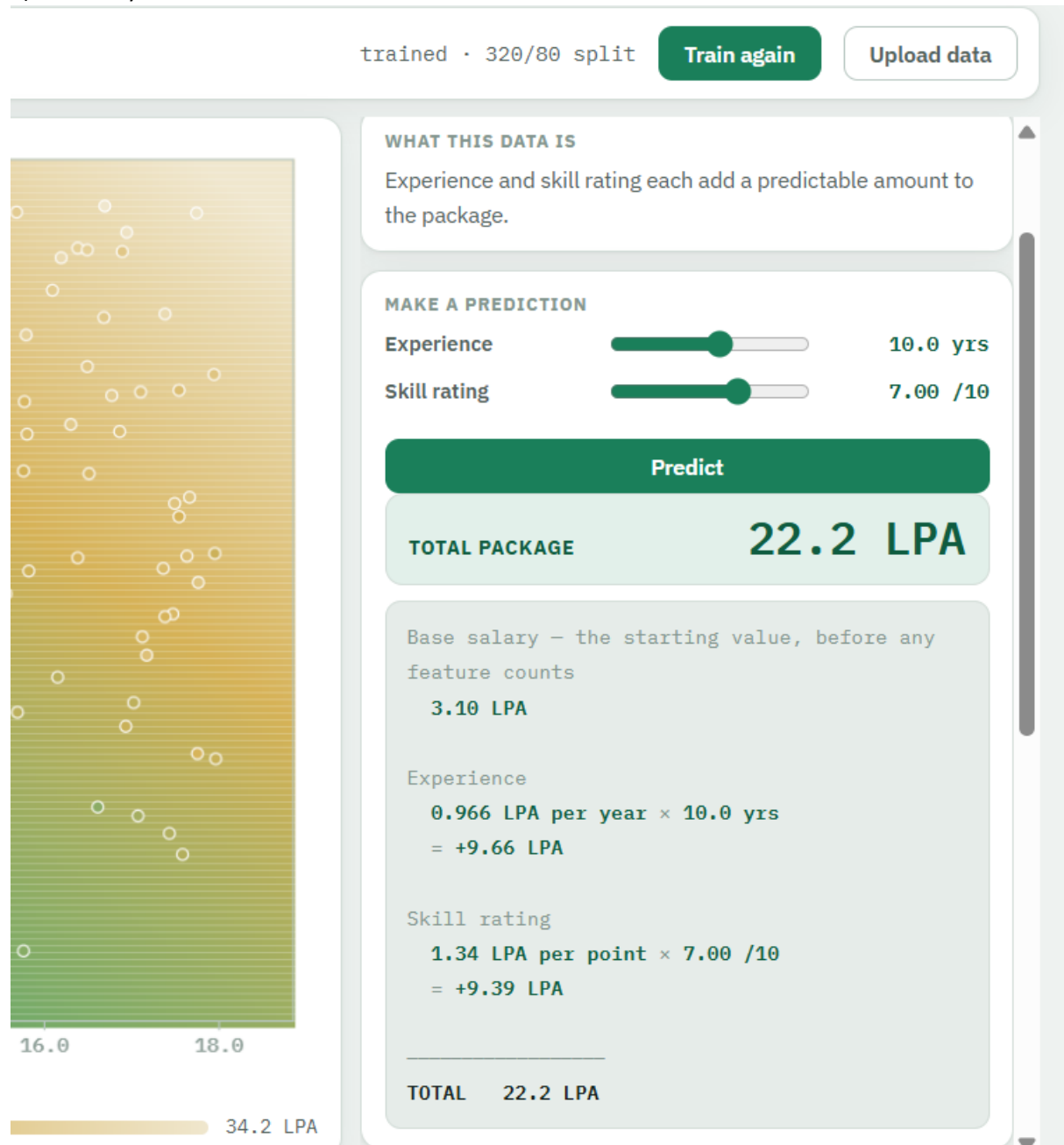
Party C $13.2 \div 15.4 = 85.9\%$

Undecided $0.919 \div 15.4 = 6.0\%$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

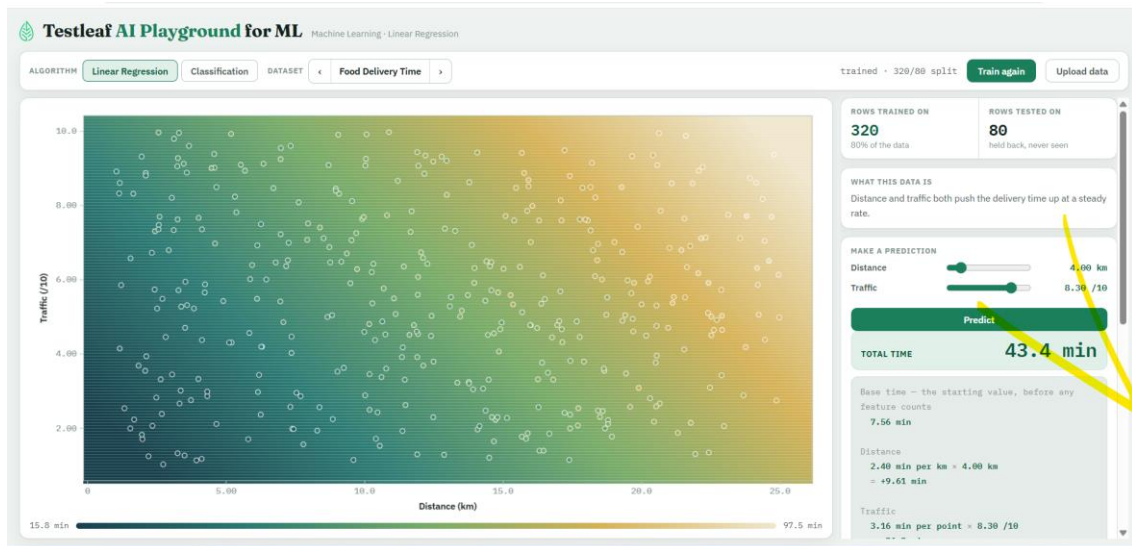
2) Linear regression

1) QA Salary



2) Food Delivery time

Check the Calculation for price



Distance 4.00 km

Traffic 8.30 /10

TOTAL TIME 43.4 min

7.56 min +

Distance

2.40 min per km × 4.00 km

= +9.61 min +

Traffic

3.16 min per point × 8.30 /10

= +26.2 min

TOTAL 43.4 min